

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

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Roll No.: 1012411011

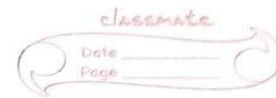
Name: Riddhee Kulkarni

Subject: Java

Assignment No.: 5

Title: Railway Reservation System

Date: 20/08/2025



Java
Assignment - 5

Name: Riddhee Kulkarni

Roll no.: 1012411011

Aim:

Write a program to demonstrate concept of user defined Packages, Wrapper classes, Abstract classes applied to railway reservation system.

Theory:

Wrapper Classes:

The wrapper class in Java provides the mechanism to convert primitive data type into objects and objects into primitive data type.

Autoboxing: Converts primitive into wrapper objects

Unboxing: Converts wrapper objects into primitive

Syntax:

```
int x = 10;
```

```
Integer y = x; //autoboxing
```

```
Integer z = 20;
```

```
int a = z; //unboxing
```

Abstract Classes

An abstract class is a class declared using the abstract keyword, that cannot be instantiated i.e cannot create an object directly. It is used when you want to provide a base class with common functionality, but also want child classes to define their own specific behaviour

Syntax:

```
abstract class Vehicle
{
    abstract void start()
    void stop()
    {
        //statement
    }
}
```

instanceof Operator

The instanceof operator is used to test whether an object is an instance of a particular class or implements an interface. It returns true or false.

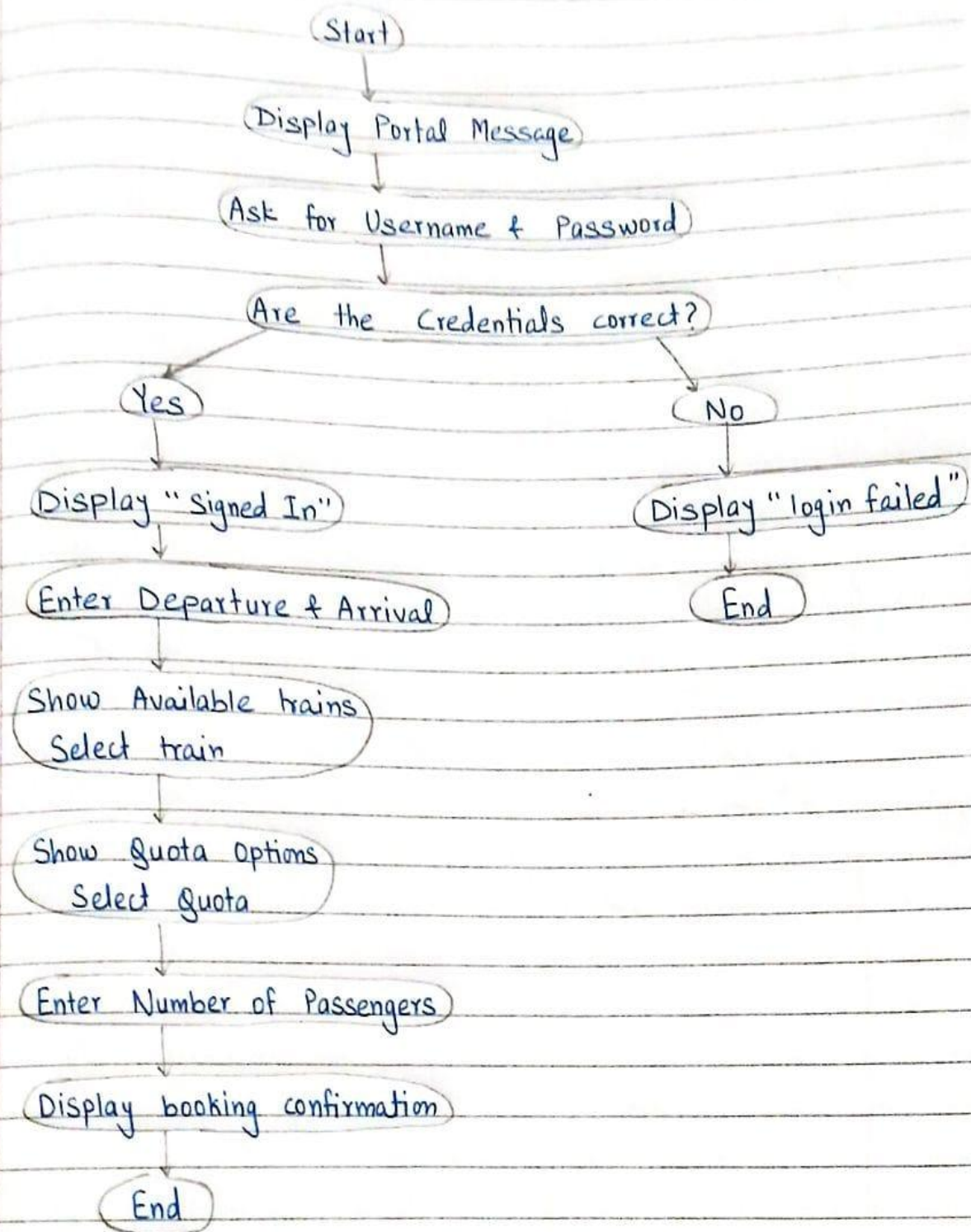
Syntax:

```
Car c = new Car();
System.out.println(c instanceof Car); // true
System.out.print(c instanceof Vehicle); // true
System.out.print(c instanceof Object); // true
```

Packages

A package is a namespace in Java, that groups related classes and interfaces together. Similar to folders on a computer, packages help organize code and avoid name conflicts. There are two types of packages, built-in package and user defined package.

Flowchart:



Package Login:

```
package login;

public abstract class Login
{
    public abstract boolean login(String username, String password);

    public void message()
    {
        System.out.println(x:"IRCTC RAILWAY BOOKING PORTAL");
    }
}

package login;

public class UserLogin extends Login
{
    @Override
    public boolean login(String username, String password)
    {
        return username.equals(anObject:"user") && password.equals(anObject:"1234");
    }
}
```

Package Booking:

```
package booking;

public abstract class Booking
{
    public abstract void destination();
    public abstract void selectTrain();
    public abstract void selectQuota();
    public abstract void numberOfPassengers();
    public abstract void bookingConfirmation();
}
```

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```
package booking;

import java.util.Scanner;

public class Ticket extends Booking {
    private Integer trainChoice, numberOfPassengers;
    private Character selectQuota;
    private String from, to, trainName;
    private Scanner sc = new Scanner(System.in);

    @Override
    public void destination() {
        System.out.print(s:"Departure: ");
        from = sc.nextLine();
        System.out.print(s:"Arrival: ");
        to = sc.nextLine();
        System.out.println("Getting results for " + from + " to " + to + "...");
    }

    @Override
    public void selectTrain() {
        System.out.println("Available Trains to " + to);
        System.out.println(x:"1. Rajdhani Express");
        System.out.println(x:"2. Shatabdi Express");
        System.out.println(x:"3. Duronto Express");
        System.out.println(x:"4. Garib Rath Express");
        System.out.println(x:"5. Humsafar Express");
        System.out.print(s:"Enter Train number (1 to 5): ");
        trainChoice = sc.nextInt();
        sc.nextLine();

        if (trainChoice < 1 || trainChoice > 5) {
            System.out.println(x:"Invalid train choice. Program will now exit.");
            System.exit(status:0);
        }

        switch (trainChoice) {
            case 1:
                trainName = "Rajdhani Express";
                break;
            case 2:
                trainName = "Shatabdi Express";
                break;
            case 3:
                trainName = "Duronto Express";
                break;
            case 4:
                trainName = "Garib Rath Express";
                break;
            case 5:
                trainName = "Humsafar Express";
                break;
        }
        System.out.println("Selected: " + trainName);
    }

    @Override
    public void selectQuota() {
        System.out.println(x:"Available quota:");
    }
}
```

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```
System.out.println(x:"a. Reservation");
System.out.println(x:"b. General");
System.out.println(x:"c. Tatkaal");
System.out.println(x:"d. Ladies");
System.out.print(s:"Enter Quota choice (a to d): ");
selectQuota = sc.next().charAt(index:0);
sc.nextLine();

if (selectQuota != 'a' && selectQuota != 'b' && selectQuota != 'c' && selectQuota != 'd') {
    System.out.println(x:"Invalid quota choice. Program will now exit.");
    System.exit(status:0);
}

switch (selectQuota) {
    case 'a':
        System.out.println(x:"Reservation");
        break;
    case 'b':
        System.out.println(x:"General");
        break;
    case 'c':
        System.out.println(x:"Tatkaal");
        break;
    case 'd':
        System.out.println(x:"Ladies");
        break;
}
}

@Override
public void numberOfPassengers() {
    System.out.print(s:"Enter number of passengers: ");
    numberOfPassengers = sc.nextInt();
    sc.nextLine();

    if (numberOfPassengers <= 0) {
        System.out.println(x:"Invalid number of passengers. Program will now exit.");
        System.exit(status:0);
    }
}

@Override
public void bookingConfirmation() {
    System.out.println(x:"\n----- Booking confirmation -----");
    System.out.println(x:"Your Booking Has Been Confirmed!");
    System.out.println(from.toUpperCase() + " to " + to.toUpperCase());
    System.out.println("Train: " + trainName.toUpperCase());
    System.out.println("Quota: " + selectQuota);
    System.out.println("Number of passengers: " + numberOfPassengers);
}
}
```

Package main:

```
package main;

import java.util.Scanner;
import login.*;
import booking.*;

public class RailwayReservationSystem {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Login step
        Login log = new UserLogin();
        log.message();

        System.out.print(s:"\nEnter Username: ");
        String username = sc.nextLine();
        System.out.print(s:"Enter Password: ");
        String password = sc.nextLine();

        if (log.login(username, password)) {
            System.out.println(x:"Signed in Successfully!");

            // Booking flow
            Ticket t = new Ticket();
            t.destination();
            System.out.println();
            t.selectTrain();
            t.selectQuota();
            t.numberOfPassengers();

            t.bookingConfirmation();
        } else {
            System.out.println(x:"Login failed!");
        }

        sc.close();
    }
}
```

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Output:

```
IRCTC RAILWAY BOOKING PORTAL
```

```
Enter Username: user
Enter Password: 1234
Signed in Successfully!
Departure: pune
Arrival: mumbai
Getting results for pune to mumbai...
```

```
Available Trains to mumbai
```

```
1. Rajdhani Express
2. Shatabdi Express
3. Duronto Express
4. Garib Rath Express
5. Humsafar Express
Enter Train number (1 to 5): 7
Invalid train choice. Program will now exit.
```

```
IRCTC RAILWAY BOOKING PORTAL
```

```
Enter Username: user
Enter Password: 1234
Signed in Successfully!
Departure: pune
Arrival: mumbai
Getting results for pune to mumbai...
```

```
Available Trains to mumbai
```

```
1. Rajdhani Express
2. Shatabdi Express
3. Duronto Express
4. Garib Rath Express
5. Humsafar Express
Enter Train number (1 to 5): 4
Selected: Garib Rath Express
Available quota:
a. Reservation
b. General
c. Tatkaal
d. Ladies
Enter Quota choice (a to d): t
Invalid quota choice. Program will now exit.
```

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IRCTC RAILWAY BOOKING PORTAL

Enter Username: user
Enter Password: 1234
Signed in Successfully!
Departure: pune
Arrival: mumbai
Getting results for pune to mumbai...

Available Trains to mumbai

1. Rajdhani Express
2. Shatabdi Express
3. Duronto Express
4. Garib Rath Express
5. Humsafar Express

Enter Train number (1 to 5): 2

Selected: Shatabdi Express

Available quota:

- a. Reservation
- b. General
- c. Tatkaal
- d. Ladies

Enter Quota choice (a to d): a

Reservation

Enter number of passengers: -1

Invalid number of passengers. Program will now exit.

IRCTC RAILWAY BOOKING PORTAL

Enter Username: user
Enter Password: 1234
Signed in Successfully!
Departure: pune
Arrival: mumbai
Getting results for pune to mumbai...

Available Trains to mumbai

1. Rajdhani Express
2. Shatabdi Express
3. Duronto Express
4. Garib Rath Express
5. Humsafar Express

Enter Train number (1 to 5): 4

Selected: Garib Rath Express

Available quota:

- a. Reservation
- b. General
- c. Tatkaal
- d. Ladies

Enter Quota choice (a to d): d

Ladies

Enter number of passengers: 3

----- Booking confirmation -----

Your Booking Has Been Confirmed!

PUNE to MUMBAI

Train: GARIB RATH EXPRESS

Quota: d

Number of passengers: 3

Conclusion:

This application demonstrates user defined Packages.
Abstract classes.